

Question:

A 3-year-old boy is brought to the physician for follow-up examination 5 days after sustaining a forehead laceration. Examination shows a linear, well-approximated laceration over the right temple. The wound is clean and dry with no exudate. There is a small amount of pink granulation tissue present. Microscopic examination of the wound is most likely to show which of the following? Answer Choices: A. Angiogenesis with type III collagen deposition B. Macrophage infiltration and fibrin clot degradation C. Capillary dilation with neutrophilic migration D. Fibroblast hyperplasia with disorganized collagen deposition

Reasoning + Answer:

<think> To answer this question, we need to consider the timing and appearance of the wound. The boy is 5 days after sustaining the injury, which is still within the early stages of the healing process. The presence of pink granulation tissue is consistent with the body's natural healing process, in which new tissue is formed to repair the damaged area. The most likely outcome of this process is the deposition of type III collagen, which is a common occurrence in the early stages of wound healing and is associated with angiogenesis. The other options are less likely to occur 5 days after the injury, as they are more consistent with later stages of the healing process or with the presence of infection. Therefore, based on the likely timing and appearance of the wound, it is most likely that the microscopic examination of the wound will show angiogenesis with type III collagen deposition. </think> <answer> Ang iogenesis with type III collagen deposition </answer>

